

Dongheon Lee

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Research Interests

My research focuses on **computer vision**, **vision-language models**, and **data-efficient learning**. I am particularly interested in:

- Few-shot learning and generalization under limited supervision
- Vision-language model adaptation and multimodal learning
- Attention mechanisms for visual recognition

Education

M.S. in Artificial Intelligence 2024 – 2026
Chung-Ang University Advisor: Joonki Paik

B.S. in Astronomy & Space Science 2016 – 2023
Kyung Hee University

Publications

C: Conference, W: Workshop, U: Under Review *: Equal Contribution, †: Corresponding Author

Journal & International Conference

- [C1] **PLuG: Pairwise Logit Gating for Expressive Attention Modulation in Vision Transformers** [Paper](#)
Dongheon Lee, Sangwoo Yoon, Seongsu Kim, and Joonki Paik[†]
IEEE Access, 2026
- [C2] **NAPP: Noise-Adaptive Prototype Perturbation for Few-Shot Learning** [Paper](#)
Il Hwan Kim, Sang Woo Yun, Dongheon Lee, Kim Seong Su, and Joonki Paik[†]
IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2026
- [C3] **CEAL: Cross-Expert Attention with Curated Peer Selection for Scalable and Privacy-Preserving Exemplar-Free Continual Learning** [Paper](#)
Seongsu Kim*, Dongheon Lee*, Sangwoo Yun, Ilhwan Kim, and Joonki Paik[†]
IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2026
- [C4] **ViSeNet: A Visual-Semantic Fusion Network for Enhancing Few-Shot Classification** [Paper](#)
Dongheon Lee, Sangwoo Yun, and Joonki Paik[†]
IEEE International Conference on Consumer Electronics (ICCE), 2026

Workshop Papers

- [W1] **Unified Visions: Multi-Modal Transformer Fusion for Comprehensive VQA Answering**
Minju Baek*, Dongheon Lee*, Jaehong Yoon and Joonki Paik[†]
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Vizwiz Workshop, 2025
- [W2] **Test-Time Augmented Ensemble for Zero-Shot Learning**
Dongheon Lee*, Minju Baek*, Jaehong Yoon and Joonki Paik[†]
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Vizwiz Workshop, 2025

Under Review

- [U1] **RoME: Routed Multi-Expert LoRA for Few-Shot CLIP Adaptation**
Dongheon Lee, Sangwoo Yoon, Seongsu Kim, and Joonki Paik[†]
- [U2] **Structured Adaptation and Subspace Regularization for Few-Shot Vision-Language Learning**
Dongheon Lee, Sangwoo Yoon, Seongsu Kim, and Joonki Paik[†]
- [U3] **STAR: Semantic-to-Temporal Anchored Refinement for Open-Vocabulary Audio-Visual Event Localization**
Dongheon Lee, Jihun Park, Yerin Lim, and Joonki Paik[†]
- [U4] **Logit-Calibrated Differential Attention for Vision Transformers**
Dongheon Lee and Joonki Paik[†]

Project Experiences

- Dronebot Combat System Response Next-Generation Intelligent Surveillance and Reconnaissance Mission Equipment** 2024 – 2026
• Funded by Korea Research Institute for Defense Technology Planning and Advancement (KRIT)
- Design of a location-based AI image interpretation platform for sewer pipe defects** 2024 – 2026
• Funded by Ministry of SMEs and Startups
- Multimodal Generative AI for Multidimensional Scene Analysis and Reproduction** 2025
• Funded by National Research Foundation (NRF)